

Week 6

COGS 108 SP 2026

DI slides!

https://github.com/shreyamusini/cogs108_sp26_di_slides_shreya

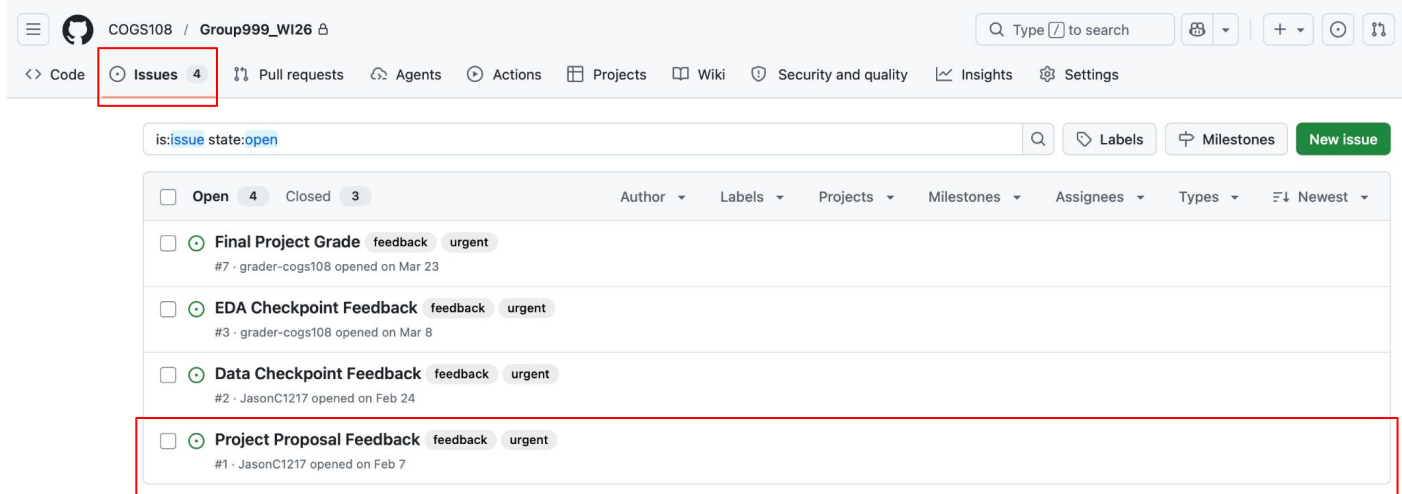
```
% git clone https://github.com/shreyamusini/cogs108_sp26_di_slides_shreya.git
```

Due dates

- A2: Wednesday (05/06)
- D5: Friday (05/08)

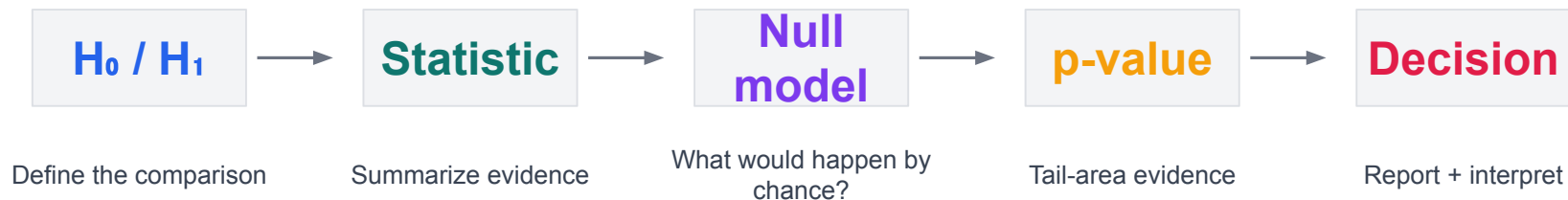
Project Proposal

- Feedback will be out by the end of today
- Make revisions in next checkpoint
- You can find your feedback in your group GitHub repository under Issues, in an issue titled "Project Proposal Feedback."



Null-hypothesis significance testing

Focus on the logic chain, not just the final p-value.



Reminder

For any scenario, try to state the null hypothesis, what statistic would be computed, what “more extreme” means, and what conclusion would be justified.

p-values, α , and practical importance

Significance and importance answer different questions.

**statistical evidence $\neq$ effect
magnitude**

**effect size + uncertainty =
interpretation**

p-value thinking

Ask: “How surprising would this statistic be if the null model were true?” Keep the conditioning clear.

Effect-size thinking

Ask: “How large is the estimated difference or relationship in the units of the problem?”

Uncertainty thinking

Ask: “How much sampling variability is around the estimate?”
Use intervals and standard errors as context.

Reminder

Practice explaining a result using all three lenses without making causal claims that the design cannot support.

p-values, α , and practical importance

Significance and importance answer different questions.

Confidence Interval

$$CI = 1 - \alpha$$

The percentage of times that a computed interval is expected to contain the true population parameter if sampling is repeated

α

The probability of being wrong for a confidence level

Interpretation

A 95% confidence interval implies that if we take 100 samples and compute 100 intervals, we expect roughly 95 of them to contain the true population mean

Choosing a two-group comparison

Start from the data structure before naming a test.

Outcome type	continuous • ordinal/rankable • categorical
Group relationship	independent samples • repeated/paired samples
Distribution shape	roughly symmetric • skewed • outlier-heavy
Variance pattern	similar spread • very different spread
Scientific target	mean difference • rank/location shift • full distribution

Reminder

In review, practice choosing the family of methods first; only then decide the exact test and interpretation.

t-test assumption audit

Parametric tests can be powerful, but only when the setup is plausible.

Before trusting a t-style comparison, inspect:

- Are observations independent within and across groups?
- Is the outcome measured on a scale where means are meaningful?
- Are the distributions approximately compatible with the model?
- Are extreme outliers or strong skew likely to dominate the mean?
- Is the spread of the groups similar enough for the chosen variant?

Compare centers only after checking the shape of the data.

Use visual diagnostics as part of the reasoning, not as decoration.

More than two groups and multiple testing

Repeated pairwise tests change the error-rate story.

When there are several groups, ask:

Global question Is there evidence of any group-level difference?

Pairwise question Which specific groups differ from each other?

Error-rate question How many comparisons are being made?

Follow-up question What correction or post-hoc plan is appropriate?

α

× comparisons

controls false positives



planned workflow

Transformations and rank-based inference

When model assumptions are strained, change the representation or the method.

Two review moves

$\log(y)$

Transformation logic

A transformation can make a skewed variable easier to model and can change how differences are interpreted.

$\text{rank}(y)$

Rank-based logic

Rank methods reduce reliance on Normal-model assumptions by analyzing ordering information rather than raw magnitudes.

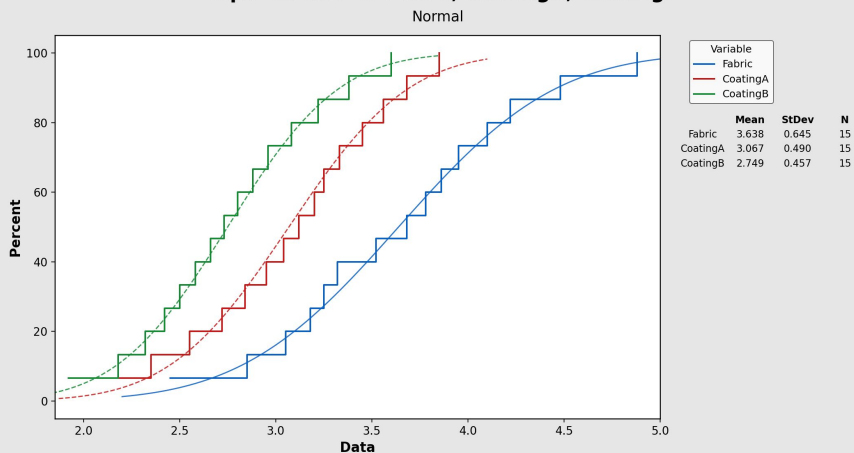
Reminder

Practice saying what information is preserved and what information is changed when you transform or rank the data.

Distribution-level comparisons

Some questions are about the whole distribution, not only the mean.

Empirical CDF of Fabric, CoatingA, CoatingB



KS-test reasoning

A distribution-comparison test asks whether two empirical CDFs look compatible with the same underlying distribution.

Score-model context

For model probability scores, ask whether the two models produce similar score distributions across users.

Reminder

Practice distinguishing “different average score” from “different score distribution.”

Spatial data: patterns and aggregation traps

Spatial summaries depend on scale, boundaries, and dependence.

Average nearest neighbor / clustering



Ask whether points are clustered, dispersed, or random — and at what scale.

MAUP

Changing geographic units can change the apparent pattern.

Ecological fallacy

A group-level pattern may not describe individuals.

Autocorrelation

Nearby or adjacent observations may not be independent.

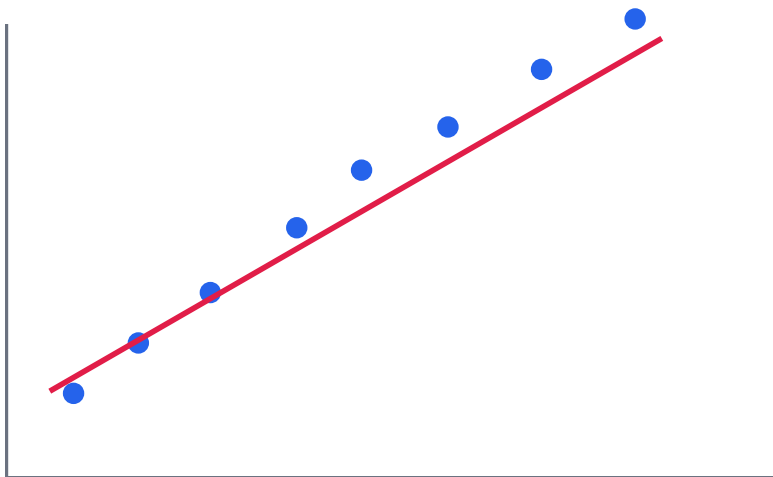
Boundary design

Political or administrative boundaries can encode incentives.

OLS regression: what the line is doing

A fitted line summarizes a conditional relationship.

$$y = \beta_0 + \beta_1 x + \varepsilon$$



β coefficient

Direction and magnitude of the estimated relationship, conditional on the model.

Standard error

Sampling uncertainty around the coefficient estimate.

Confidence band

Uncertainty around the fitted mean relationship, not the full range of outcomes.

Model fit versus model complexity

Adding predictors changes both fit and penalty-based metrics.

R^2

Raw fit

How much variation is explained by the fitted values?

adjusted R^2

Fit with a predictor penalty

Does the new predictor improve the model enough to justify its complexity?

BIC

Information criterion

Compare candidate models while accounting for likelihood and parameter count.

Reminder

Practice predicting the direction each metric might move when you add useful versus irrelevant predictors.

OLS diagnostics and failure modes

A good-looking line is not enough.

Linearity

Mean relationship is represented by the model form.

Independence

Residuals are not correlated across observations.

Homoscedasticity

Residual spread is roughly stable across fitted values.

Collinearity

Predictors do not duplicate each other's information.

Time series caution

Sequential observations may carry memory from prior time points.

Condition number

diagnostic clue, not a conclusion by itself

Quiz Time

NO Phone, NO Communication, No Notes

Access Code:

Your time to ...

- Work on projects
- Finish A2
- Finish D5
- Have fun...